

Special Topics in Health and Exercise Science: Sports Performance Analytics

HES 3000-002

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About me

- Graduate Instructor
 - HES 3000-002: Sports Performance Analytics
 - HES 3883: Principals of Endurance Training
 - HES 2131, 2913, 2823
- Background & Research
 - Endurance sports and physiology
 - Performance Analytics
 - Athlete monitoring and training load management



Why Performance Analytics?

Bring together physiology, data and real decisions

- Improve performance by:
 - understanding how athletes respond to training.
 - flag concerning patterns before they become larger performance or health problems.
 - guide better decisions with information.

Tell me about yourself!

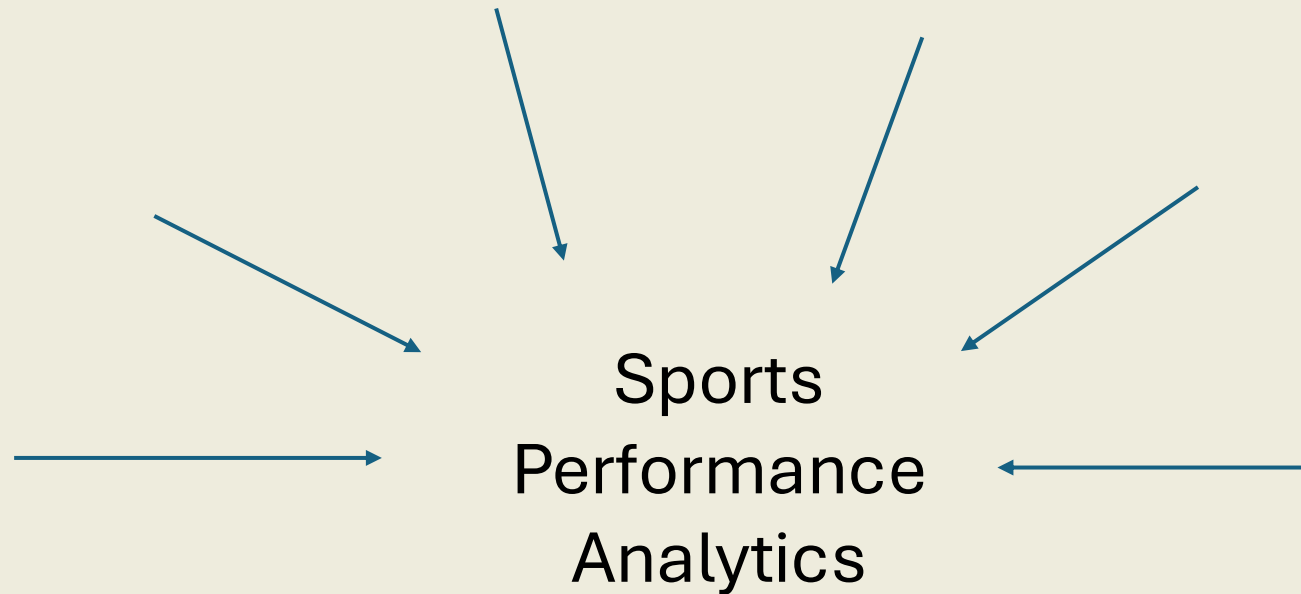
In your index cards:

- Name
- Year + Major
- Favorite sport / team
- One thing you're hoping to learning in this class

Components of Performance Analytics

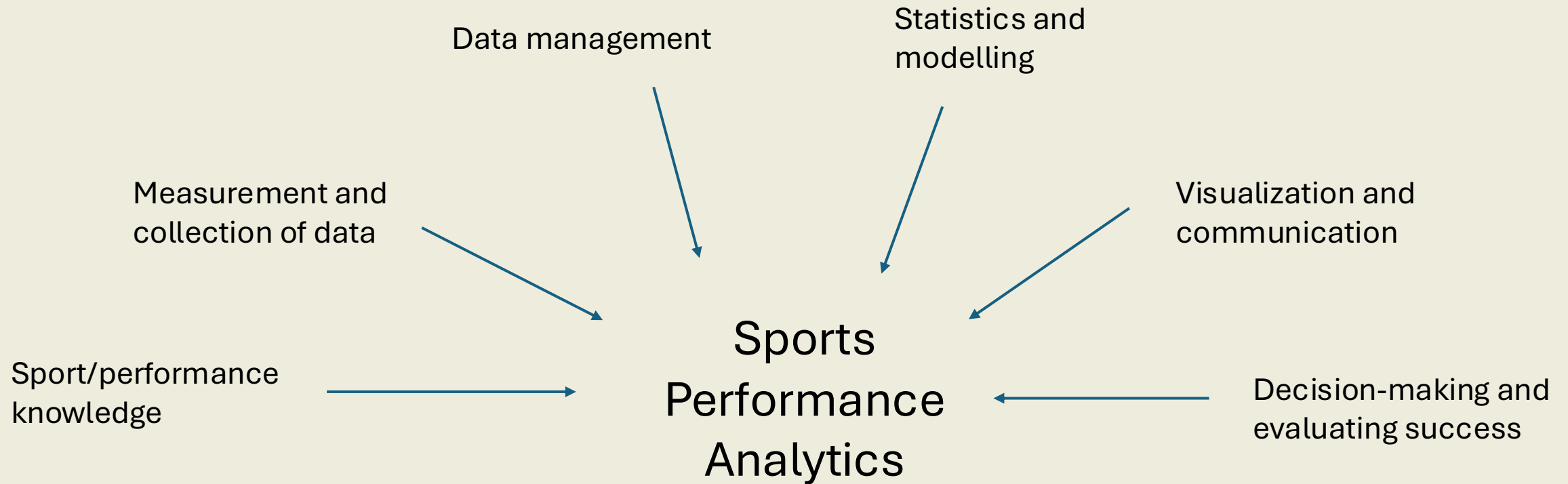
Performance analytics is not a single skillet

What skills do we need?



Components of Performance Analytics

Performance analytics is not a single skillset



What skill is the most important?

Working Definition for This Course

- **Sports Performance Analytics:**
 - Management of structured **performance data**
 - **Application** of analytic models
 - Inform **decision makers**
 - Gaining a **competitive advantage**

Skills to Build during this semester

- **Analytics foundations:**
 - What sport analytics is and formulating good questions we can answer
- **Data:**
 - Sources, structure, quality, validity, reliability, and ethics
- **Describe data:**
 - Center, spread, distributions, and athlete profiling
- **Visualize data:**
 - Showing patterns clearly and honestly
- **Explore relationships:**
 - Association, regression, and interpretation
- **Making decisions:**
 - Thresholds, probabilities, models, and communication
- **Application:**
 - Example applications + semester-long group project

The Scope of this Course

- We are interested in questions related to:
 - Scouting, recruiting, and roster decisions
 - Athlete performance and training
 - Strategy and competition
- We are **not** working on sports business problems
 - Ticketing
 - Marketing and sponsorship
 - Operations

Brainstorming Activity

List a few example of data or analytics being used in sports.

In pairs

1. Give each person an opportunity to share.
2. Select one or two ideas to discuss further.
3. For each selected idea, consider:
 - What information was being collected or presented?
 - What was it being used for? To understand? To decide?
 - Did you find it useful, convincing, or questionable?

Brainstorming Activity

- The examples generated today involve different:
 - Sports and competitive settings
 - Types of data
 - Analytical approaches
 - Audiences and decisions
- Throughout the semester, we will develop a **systematic** method to understand and/or investigate.

Syllabus

Assessments

- 12 quizzes:
 - Review the previous week's material; the two lowest scores are dropped.
- In-class activities:
 - Apply concepts to sports problems and earn participation credit.
 - > 80% completion = 10 pts, 60-79% completion = 5 pts, < 60% completion = 0.
- 4 case studies:
 - Interpret data, evaluate methods, visualize findings, and reflect on analytical choices.
- 2 exam:
 - Demonstrate your understanding of concepts, applications, and the final project.

Assessments

Evaluation	Points
Quizzes (10)	100
Attendance	10
Case study (4)	100
Final project	100
Exams (2)	200
Total	510

Course policies

- Attendance
 - In-class activities count toward attendance and cannot be made up.
 - Grade rounding at the end of the semester is only applied if full attendance points is earned.*
- Deadlines:
 - Late quizzes and exams are not accepted.
 - Other late work loses 10% per day unless arranged in advance.
- AI:

*Check syllabus for full details

Resources

- Canvas
 - All materials used in this course will be available on Canvas
 - Class-wide communications will be done via Canvas Announcements
- Communication
 - Email: saryu@ou.edu (**DO NOT** message me on Canvas)
 - Office: SJSC 103 on Tuesdays at 1:30-2:30 or by appointment
 - *Help me help you* - **please** reach out if you need something from me!
- Reasonable accommodation policy
 - All accommodations are facilitated through the Accessibility and Disability Resource Center.
 - If you require accommodations, contact adrc@ou.edu or (405)2325-3852.

Looking Ahead

- **Next Class:** Evolution of sports analytics
- **Before Next Class:**
 - Read the syllabus
 - Get familiar with Moneyball, by Michael Lewis